

Chen Xie

Department of Physics & Astronomy

The Johns Hopkins University

3701 San Martin Drive, Baltimore, MD, 21218, USA

Phone: +1 443 925 8789

Email: cx@jhu.edu

Website: chenxie.fr

Research Position

- **Assistant Research Scientist** Baltimore, MD
The Johns Hopkins University 2023 - present
- **Graduate Student Researcher** Marseille, France
Aix-Marseille Université 2020 - 2023
- **Summer Student internship** Dwingeloo, The Netherlands
ASTRON/JIVE 2017

Education

- **Aix-Marseille Université** Marseille, France
PhD in Astronomy awarded in Feb 2024
 - PhD Thesis: Development of advanced post-processing methods for the direct detection of exoplanets with ground-based high-contrast imagers
 - Thesis Advisor: Dr. Arthur Vigan, Dr. Elodie Choquet
- **Leiden University** Leiden, The Netherlands
M.Sc. in Astronomy Feb 2020
 - Major Thesis: VLT/MUSE as a high-contrast imager
 - Thesis Advisors: Prof. Matthew Kenworthy, Dr. Jos de Boer, and Dr. Sebastiaan Haffert
 - Minor Thesis: Radio Observations of Frontier Fields Clusters Abell S1063 and Abell 370: The Discovery of New Radio Halos
 - Thesis Advisor: Prof. Reinout van Weeren
- **Xiamen University** Xiamen, China
B.Sc. in Physics June 2016
 - Bachelor Thesis: On the Host Galaxy of GRB 150101B
 - Thesis Advisor: Prof. Taotao Fang

Selected Grants & Awards

Total awards for JWST programs (US-admin PI), **\$149,369** Cycle 3 – 4
Poster Prize, Lyot conference 2022
Oort Scholarship (**€22k + tuition waiver**), Leiden University 2018 – 2020
First Prize of Lin Qiao Prize for Excellent Undergraduate Research Project, Peking University 2016

Mentoring

5. Chen, J.-Y. (MSc student, Xiamen Univ., ongoing) – SAFFORN survey, spiral motion measurement
4. Zhong, H. (PhD student, Sun Yat-sen Univ., ongoing) – Reflectance spectra of HD 32297 debris disk; paper in prep.
3. Xie, C.-Y. (PhD student, Univ. of Arizona, 2024) – publication in *Universe*, 10, 465 (2024).
2. Ahmad, Y. (MSc student, EuroPhotonics, 2022–23) – developing image-processing techniques
1. Rahman, S. (MSc student, EuroPhotonics, 2021–22) – developing image-processing techniques

Selected Publications (128 citations from 9 first-/corresponding-author papers)

1st-/Corresponding-authored:

12. **Xie, C.**, et al. *Water ice in the debris disk around HD 181327*, **Nature**, 641, 608 (2025).
Research Highlight: water ice discovery
11. **Xie, C.**, et al. *Reference-star differential imaging on SPHERE/IRDIS*, **A&A**, 666, A32 (2022).
Research Highlight: archival RDI with 2× improvement
10. **Xie, C.**, et al. *Searching for proto-planets with MUSE*, **A&A**, 644, A149 (2020).
Research Highlight: first MUSE survey
9. **Xie, C.**, et al. *DESTINYs: Dynamical evidence of a spiral-arm-driving protoplanet*, **Universe**, 10, 465 (2024).
8. Ren, B.B., **Xie, C.**, et al. *A companion in V1247 Ori supported by spiral motion*, **A&A**, 681, L2 (2024).
7. **Xie, C.**, et al. *Dynamical detection of a companion driving a spiral arm*, **A&A**, 675, L1 (2023).
6. **Xie, C.**, et al. *MUSE view of asymmetric jet from HD 163296*, **A&A**, 650, L6 (2021).
5. **Xie, C.**, et al. *The discovery of radio halos in the Frontier Fields clusters Abell S1063 and Abell 370*, **A&A**, 636, A3 (2020)
4. **Xie, C.**, et al. *On the Host Galaxy of GRB 150101B and the Associated Active Galactic Nucleus*, **ApJL**, 824, L17 (2016)

2nd/3rd authored:

3. Betti, S.K., Chen, C.H., **Xie, C.**, et al. *Detection of ices in the debris disk around β Pictoris*, **Science** (under review since Feb. 2025).
2. Matthews, E.C., Bonnefoy, M., **Xie, C.**, et al. *First scattered-light images of HD 112810 debris disk*, **A&A**, 679, A58 (2023).
1. Uyama, T., **Xie, C.**, et al. *Keck/OSIRIS Pa β constraints on PDS 70b*, **AJ**, 162, 214 (2021).

Leadership

Spiral motion lead, VLT/SPHERE SAFFRON survey (PI: B. Ren; 69.5 hrs; 40+ members), primary goal

Selected Meetings since 2022

Gordon Research Conferences, Origins of Solar Systems, Poster, South Hadley, MA, 2025
National Capital Area Disks Meeting, **Talk**, Washington, DC, 2025
STScI JWST science conversation, **Invited talk**, Baltimore, MD, 2025
STScI-JHU ExoJamboree, **Talk**, Baltimore, MD, 2024
Dust Devils workshop, **Talk**, Tucson, AZ, 2024
ETH Zürich Exoplanets & Habitability Seminar, **Invited talk**, Zürich, 2022
Debris discs: At Home and Abroad, **Talk**, Jena, 2022
Spirit of Lyot 2022, Poster, Leiden, 2022

Service

Member, Roman Coronagraph Community Participation Program 2025 –
Referee, AAS journals 2024 –
Referee, A&A 2023 –